

Technical Service Report & Recommendation – CCTV Recording Storage

Client: [Gulf Aluminium]
Site Location: [Asker]
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1 . Purpose of Inspection

A technical inspection was conducted to evaluate whether the existing CCTV recording system complies with the Ministry requirement of maintaining 120 days video recording retention under motion recording mode.

The inspection included verification of the DVR model specifications, available HDD storage capacity, and system limitations.

2. Existing System Evaluation

DVR Unit 1

Model: DH-XVR5232AN-I3

Manufacturer: Dahua Technology

Technical Findings:

- Maximum supported HDD storage: 32 TB
- Current camera count connected: 25 Cameras
- Storage required for 120 days motion recording: Approx. 34 TB

Technical Limitation Identified:

- The DVR supports a maximum of 32 TB only
- The device does not have additional SATA ports to increase the storage capacity
- HDD capacity cannot be expanded beyond the device limitation

Result:

This DVR cannot meet the 120-day recording retention requirement due to hardware storage limitations.

DVR Unit 2

Model: DHI-XVR5232AN-S2

Manufacturer: Dahua Technology

Technical Findings:

- Maximum supported HDD storage: 16 TB
- Number of SATA slots: 2 slots only

- Storage required for 120 days motion recording: Approx. 42 TB
- Technical Limitation Identified:
- The DVR supports very limited HDD capacity
 - Only two SATA slots are available
 - The required storage capacity exceeds the maximum supported storage

Result:

This DVR cannot achieve the required video retention period due to insufficient storage capability.

3. Compliance Concern

Based on the above findings, the existing DVR systems are technically incapable of supporting the required 120-day recording period specified by the Ministry.
Even with maximum possible HDD installation, the system will still fail to meet the required storage duration due to hardware limitations.

4. Recommended Technical Solution

To ensure compliance with the Ministry's CCTV recording retention requirement, we strongly recommend replacing the existing DVRs with the following upgraded model:

Recommended Model:DH-XVR5832S-I3

Manufacturer: Dahua Technology

Key Technical Advantages:

- 8 SATA HDD slots for higher storage expansion
- Supports up to 16 TB per HDD
- Maximum storage capacity up to 128 TB
- Suitable for large camera systems
- Capable of achieving 120 days recording retention
- Allows future storage expansion if additional cameras are added

5. Proposed Scope of Work

The following works are recommended to upgrade the system:

1. Supply of new DVR model DH-XVR5832S-I3
2. Removal of existing DVR units
3. Installation and configuration of new DVR
4. Installation of required HDD capacity to meet 120 days recording
5. System configuration and optimization for motion recording
6. Full testing and verification of recording retention
7. Integration with existing CCTV cameras and network

6. Final Recommendation

After detailed technical evaluation, it is confirmed that the existing DVR models cannot be upgraded to meet the required recording storage capacity.

Therefore, replacement of the current DVR systems with the DH-XVR5832S-I3 model is technically necessary to ensure:

- Compliance with Ministry regulations
- Reliable long-term video storage
- System scalability and stability

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